Luis Lopez Rondon

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Education

University of South Florida — **Tampa, FL**
Bachelor of Science in Electrical Engineering — ***GPA: 4.00*** | *May 2029*

Hillsborough College — **Tampa, FL**
Associate of Arts — ***GPA: 4.00*** | *May 2026*

Relevant Coursework: Differential Equations, Digital Circuits

Honors: National Merit Finalist, Sunshine State Scholar, Judy Genshaft Honors College

Technical Skills

Hardware: PCB design, buck/boost converter PCBs, Li-ion battery pack design, soldering (SMD, THT)
Embedded: STM32 (CubeIDE, CubeMX), ESP32, Teensy | I2C, SPI, UART, CAN, I2S, USB
EDA & Simulation: Altium Designer, KiCad, LTspice, Keysight ADS
Programming: C/C++, MATLAB
Spoken Languages: English, Spanish

Engineering Experience

IEEE at USF EXO — **Tampa, FL**
Electrical Systems Lead, Lower-Limb Exoskeleton — *Aug 2026 – Present*

- Leading electrical development of a **unilateral exoskeleton** with two **48V CubeMars AK80-9** actuators for hip and knee assistance and a passive ankle mechanism.
- Designing the power system: DC-DC rails for the Jetson Orin Nano and Teensy 4.1, and a relay-based **hardware e-stop** that cuts the motor bus while keeping logic powered.
- Modeled actuator torque and current in **MATLAB** from OpenSim-derived gait data: **±15 N·m** per-joint envelope, **19.49A** peak, **44.3%/62.9%** hip/knee torque-weighted coverage.
- Sized the 48V Li-ion pack from modeled battery draw (**∼10A peak, 3–4A mean** vs. a **>50A** initial estimate) and ran a weighted benefit analysis of pre-built vs. custom packs.
- Developing gait-control and verification procedures using joint encoders, IMUs, and heel/toe force sensors to time assistance across gait-phase transitions.

Projects

PID-Stabilized Parkinson's Spoon | *IEEE at USF EXO* — *Aug 2026 – Present*

- Designed the PCB for a PID-controlled Parkinson's spoon (**STM32G431**, LSM6DSV16X IMU) with 1S LiPo, TP4056 charging, DW01A/FS8205 protection, and a custom **5V servo boost converter**.
- Taught the design in an Altium workshop, taking **∼10** EXO members through their first PCB layout.

Mime: Mimetic Robotic Hand | *IEEE at USF EXO* — *May 2026 – Aug 2026*

- Designed an **ESP32-S3** + **PCA9685** PCB in Altium for a hand driven by a flex glove and OpenCV.
- Drove **5** MG996R servos from a dedicated **6V** supply, kept separate from the AMS1117 logic rail.

Audio Spectrum Analyzer | *Teensy 4.0, SGTL5000, C++* — *Aug 2026 – Present*

- Built a real-time analyzer on a **Teensy 4.0** + **SGTL5000 codec** (I2S audio, I2C control) with USB/line-in input, a **1024-point** FFT in **16 log-spaced bands**, and an ILI9341 SPI TFT.
- Consolidating the design onto a custom **STM32**-based PCB with analog filtering on the audio input.